

USOPC Data Quality Audit — Phase 2

Overview

The Master Data Management (MDM) audit was conducted to evaluate the consistency and reliability of athlete demographic information across the USOPC's source systems and the consolidated golden record. The analysis examined missing, invalid, and inconsistent values across multiple fields to identify data-quality risks that affect reporting accuracy and downstream analytics.

Overall, the audit revealed that while the MDM layer successfully integrates data from multiple sources, several upstream systems still introduce formatting, validation, and completeness issues that compromise matching accuracy and record reliability.

Key Findings

1. **Duplicate MDM Person IDs:**
Five IDs were entered three times and 106 IDs appeared twice, suggesting duplicate creation or weak match-merge logic.
 2. **Phone and Country Fields:**
Combined missing or invalid rate near 50 %. Many phone numbers contained incomplete digits or non-numeric symbols; country values were inconsistent or misspelled.
 3. **Email, Gender, and Birthdate:**
Low invalid rates (< 5 %), but minor discrepancies remained across certain systems.
 4. **Sport Codes:**
Three codes lacked proper confirmation (COV, BLD_P, STA).
 5. **Golden Record Integrity:**
The MDM layer consolidated data effectively but inherited upstream invalid values, highlighting the need for cleansing and validation prior to ingestion.
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Root Causes

- Non-standardized field formats across systems (especially Date and Phone).
 - Lack of automated validation at the point of entry.
 - Manual data entry errors and incomplete records.
 - Misaligned update schedules and synchronization delays between systems.
 - Absence of centralized data stewardship to enforce consistent rules and ownership.
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Recommendations

♦ Quick Wins (0 – 3 Months)

<u>Focus Area</u>	<u>Recommendation</u>	<u>Expected Impact</u>
Birthdate	Enforce a single date format (YYYY-MM-DD) and auto-flag implausible or future dates.	Reduce invalid birthdates.
Phone	Require numeric-only 10-digit input; strip symbols during ingestion; allow flexibility for non-US formats.	Eliminate most phone-format errors.
Email	Apply regex validation and require completion before saving a record.	Improve contact integrity and reduce rework.
Gender	Restrict accepted values to “Male,” “Female,” or “Prefer Not to Say.”	Remove ambiguous text entries.
Source Systems	Run monthly pre-MDM data-quality checks by each system analyst.	Prevent upstream errors from entering MDM.
Sport Code	Re-validate sport-code list; correct unconfirmed codes (COV, BLD_P, STA).	Improve accuracy and standardization of sport classifications.

♦ Long-Term Improvements (3 – 12 Months)

<u>Focus Area</u>	<u>Recommendation</u>	<u>Expected Outcome</u>
MDM Matching Logic	Add pre-match data-cleansing and weighted confidence scoring to reduce duplicates.	More accurate golden record consolidation.
Data Governance	Establish a USOPC Data Stewardship Council to define and maintain data standards and validation rules.	Sustained cross-system consistency.
Automation	Integrate automated data-quality checks within ETL/Tableau Prep flows and nightly MDM loads; route exceptions to analysts.	Early detection of data anomalies.
Standardization	Publish a shared data dictionary and enforce standardized field formats across all systems.	Single, authoritative version of truth.

Communication & Training	Host quarterly analyst syncs and publish short data-quality playbooks per field.	Stronger collaboration and accountability.
Country Validation	Standardize “USA” country code; correct inconsistent spellings (e.g., USAA, AUSAtrai).	Improved accuracy for athlete nationality.
Completeness Rules	Define mandatory fields for each source system and enforce them at entry.	Increased completeness and reliability of key fields.

Next Steps

1. Identify and prioritize key fields (e.g., Birthdate, Phone, Country) for immediate remediation.
 2. Implement standardized validation logic in all systems by Q1 next year.
 3. Track key performance indicators (% Missing, % Invalid) monthly using the Tableau dashboard as a live audit tool.
 4. Extend the audit framework to performance and training data once demographic quality stabilizes.
 5. Use the MDM Golden Record as the authoritative source to propagate corrections back to contributing systems, ensuring alignment and completeness.
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Conclusion

The audit confirms that most data-quality issues originate from inconsistent entry practices and missing validation controls within source systems, not from the MDM process itself.

By enforcing standardized formats, automating validation, and introducing a unified governance model, the USOPC can substantially improve athlete-record accuracy and strengthen trust in its golden record. Ongoing collaboration between data stewards and system analysts will sustain these improvements and enable faster, more confident decision-making across the organization.